

Private-Label Coffee Strategy

Designing a data-led coffee product launch for a UK grocery retailer

Marketing Analytics | Retail Strategy | Power BI | R

KEY FINDING A sustainable dark-roast 100% Arabica capsule coffee priced at £5 offers the strongest balance of convenience, quality perception, sustainability appeal, and private-label value.			
4,961 cleaned transaction rows	922 customers analysed	98.5% target segment coffee buyer rate	32.0% predicted market share

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Private-label coffee launch at a glance

This portfolio case study reframes the original marketing analytics assignment into an employer-facing business case. The focus is not only on the methods used, but on how the analysis supports a practical product launch decision.

Business question	Which private-label coffee product should a UK grocery retailer launch, and which customers should it target?
Dataset	4,961 cleaned transaction rows, 922 customers, 10 product categories, 200 conjoint respondents and 24 competitor products.
Target segment	High-value coffee loyalists - 135 customers, 14.6% of the base, 98.5% coffee buyer rate and £297 average customer value.
Recommended product	Capsule coffee priced at £5, dark roast, 100% Arabica blend and sustainability claim.
Commercial evidence	Predicted 32.0% market share versus three competitor alternatives, supported by conjoint utility and PCA positioning analysis.

// 02 BUSINESS QUESTION

Turning customer and market data into a product launch decision

UK grocery retailers operate in a competitive private-label market where own-brand products increasingly compete on quality, differentiation, sustainability and value. The challenge for the retailer is not just to launch a coffee product, but to select the right target segment, product attributes, price point and market position.

This project asks: how can customer transaction data, product preference data and competitor positioning evidence be used to design a private-label coffee product with strong customer and market fit?

// 03 DATA AND ANALYTICAL APPROACH

How the analysis was built

The analysis combined customer-level behavioural segmentation with product preference modelling and market positioning. This creates a direct link between who to target, what to launch, where to position the offer and how to communicate it.

Data preparation Cleaned transaction data, removed duplicate and incomplete records, created TotalSpend and reviewed one highly influential coffee transaction.	Customer analytics Used EDA, RFM scoring, hierarchical clustering, K-means clustering, ANOVA and LDA to identify and profile customer groups.
Product analytics Used conjoint analysis to estimate attribute importance, part-worth utilities, willingness to pay and predicted market share.	Positioning analytics Used PCA to map competitor products and project the proposed private-label product into the market space.

What the analysis showed

Insight 1: Coffee is a credible launch category Coffee generated £11,233 in sales, ranking second across the product categories. However, only 39.4% of customers bought coffee, so the opportunity depends on targeting the right customers.	Insight 2: The target is behaviourally defined High-value coffee loyalists combine high purchase frequency, low recency, broad basket value and a 98.5% coffee buyer rate. Demographics alone did not identify this group reliably.
Insight 3: Convenience dominates product choice Conjoint analysis showed format was the largest preference driver at 44.5%, followed by price at 31.1%. Capsule format had the strongest positive utility.	Insight 4: The proposed product has a defensible position The £5 sustainable dark-roast Arabica capsule product achieved the highest predicted market share at 32.0%, ahead of the nearest competitor at 26.7%.

Evidence behind the recommendation

The strongest evidence comes from three connected outputs: the target segment profile, the product preference analysis and the market positioning analysis.

Target segment evidence

Segment	Customers	Avg value	Coffee buyer rate	Business interpretation
High-value coffee loyalists	135	£297	98.5%	Primary launch target: high value and high category relevance.
Committed coffee buyers	221	£80	100.0%	Secondary activation group: strong coffee relevance but lower value.
General grocery shoppers	297	£102	1.7%	Low coffee fit despite wider grocery activity.
Low-value inactive customers	269	£16	1.5%	Low priority for launch targeting.

Business takeaway: the launch should focus on behaviourally identified coffee shoppers rather than broad demographic targeting. LDA results showed that income, age and household size were not sufficient to reliably classify the primary target segment.

Power BI view of the launch strategy

The dashboard summarises the evidence for the recommended product, price, positioning and target audience. It connects the analytical outputs to the 4P marketing mix.

Private-Label Coffee Launch Strategy Dashboard

Evidence-based marketing mix for the proposed PB coffee product

High-Value Coffee Loyalists
Target segment

98.52%
Target Coffee Buyer Rate

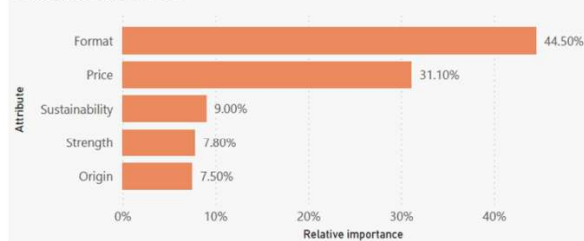
£297
Avg customer value

Capsule | £5 | Dark | Arabica | Sustainable
Product Concept

Accessible Premium
Positioning

Product Design Evidence

Product choice drivers



Recommended product

Capsule format
£5 price point
Dark roast
100% Arabica blend
Sustainability claim

Insight:

Format Dominates Choice, Supporting a Capsule-led launch.

Price: Accessible Premium Evidence

Average pack price by brand

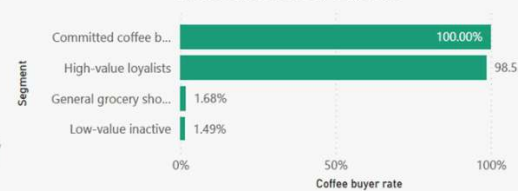


Insight:

£5 supports a premium private-label position without moving into the highest branded price tier.

Place & Promotion Targeting Evidence

Coffee buyer rate by segment



Average customer value by segment



Place recommendation

In-Store:

Coffee aisle
Near Capsule Competitors
Nescafe, Starbucks and Costa

Online:

Coffee Category Banners
Personalised recommendations

Promotion recommendation

Primary audience:

High-value coffee loyalists

Secondary audience:

Committed coffee buyers

Channels:

Loyalty-card coupons
App offers
Email Offers

Figure 10. Power BI dashboard: private-label coffee launch strategy evidence for the 4P mix

Figure 1. Power BI dashboard summarising product design, price, place and promotion evidence.

Conjoint analysis and product concept

The conjoint results provide a clear product design signal. Format accounted for 44.5% of preference variation, followed by price at 31.1%. Sustainability added a smaller but meaningful preference contribution at 9.0%.

Evidence	Metric	Result	Decision value
Attribute importance	Format	44.5%	Capsule-led product design is justified.
Attribute importance	Price	31.1%	Price remains important, so £5 must signal value.
WTP estimate	Capsule format	£3.05	Convenience creates meaningful customer value.
WTP estimate	Sustainability claim	£0.76	Sustainability supports premium private-label positioning.
Market simulation	Proposed PB product	32.0%	Highest predicted share among the tested alternatives.

Positioning evidence

PCA positioned the proposed private-label product in a moderate-convenience space between mass convenience brands and traditional ethical products. This supports an accessible premium position rather than a low-cost own-label position.

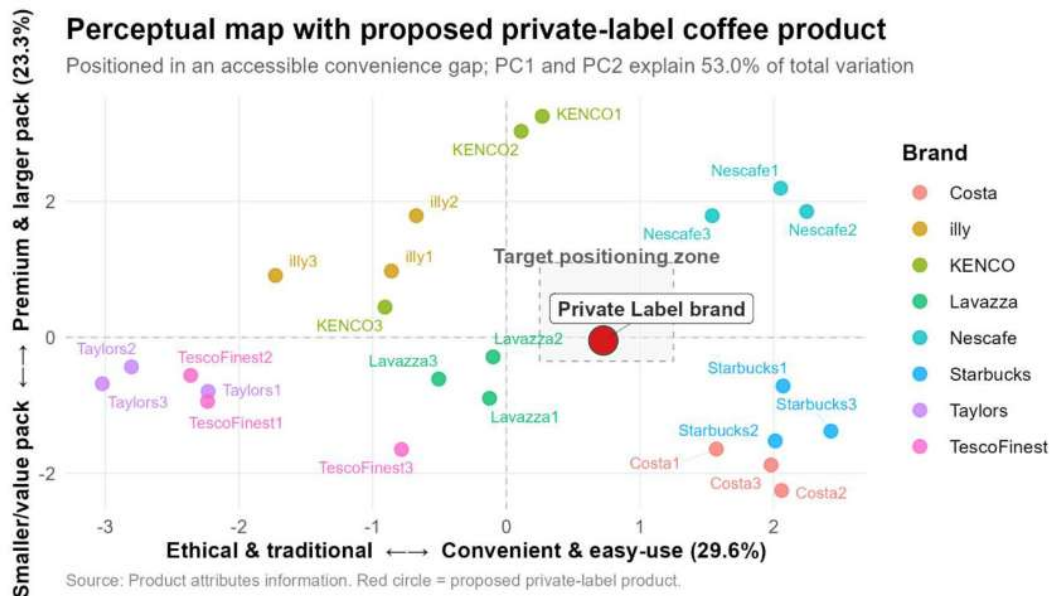


Figure 2. Perceptual map with proposed private-label coffee product.

What the analysis helps the retailer decide

This project does not claim that a real product was launched or that sales increased in practice. Its value is analytical and decision-focused. It shows how marketing analytics can turn transaction data, preference data and competitor data into a structured retail product recommendation.

Targeting decision Prioritise High-value coffee loyalists as the primary audience and use Committed coffee buyers as a secondary trial audience.	Product decision Launch a capsule format product because convenience is the dominant purchase driver.
Pricing decision Use a £5 price point to balance accessible private-label value with a premium quality signal.	Positioning decision Position the product as accessible premium: convenient, sustainable and quality-led, without competing directly with the highest priced brands.

// 07 RECOMMENDATIONS

Recommended 4P marketing strategy

4P Element	Recommendation	Analytical evidence
Product	Launch a sustainable dark-roast 100% Arabica capsule coffee.	Format was the largest conjoint driver at 44.5%; sustainability added £0.76 mean WTP.
Price	Set the price at £5 as an accessible premium private-label position.	£5 sits below premium brands such as Lavazza and illy while still signalling quality.
Place	Prioritise the coffee aisle, online grocery platform and personalised recommendations.	Target customers are best reached through behavioural and loyalty data, not demographic-only classification.
Promotion	Promote convenience, quality, sustainability and private-label value.	The positioning gap sits between convenience brands and ethical/traditional products.

// 08 LIMITATIONS AND NEXT STEPS

How the analysis should be interpreted

The analysis provides a strong decision framework, but it should be treated as a product strategy case study rather than a confirmed commercial launch result.

Limitation: simulated product performance The proposed product does not yet exist, so market share and positioning outputs are scenario-based rather than observed market outcomes.	Limitation: average preference effects Conjoint results use average utilities and may hide meaningful differences between individual consumers.
Next step: live market test Run a limited in-store or online trial to validate demand, price acceptance and repeat purchase behaviour.	Next step: loyalty-data activation Use loyalty-card data to target coffee loyalists and measure private-label switching.

// 09 PORTFOLIO REFLECTION

What this project demonstrates

This project demonstrates my ability to connect marketing theory, customer analytics, statistical modelling and dashboard communication into a practical retail recommendation. The strongest part of the project is the link between

segmentation, product design and positioning: the target customer, product attributes, price point and promotional logic are all supported by analysis rather than assumption.

It also reflects an important business analytics lesson: dashboards and models are most valuable when they help stakeholders make clearer decisions. In this case, the analytical outputs support a clear launch decision, a defined target segment and a coherent 4P strategy.

// APPENDIX

Tools and methods summary

R	Data preparation, EDA, RFM, clustering, LDA, conjoint analysis, PCA and visualisation.
Power BI	Dashboard design and stakeholder-facing communication of product strategy evidence.
Segmentation	RFM, hierarchical clustering and K-means used to identify customer groups.
Conjoint analysis	Used to estimate attribute utilities, willingness to pay and predicted market share.
PCA	Used to map competitor products and position the proposed private-label coffee concept.